



Introduction to Statistics and Probability

Assignment # 1

2025-2026

Exercise 1 – Prove that for a given sample space Ω , the collection of all subset of Ω is a σ -algebra.

Exercise 2 – Imagine a population of $N + 1$ urns. Urn number k contains k red balls and $N - k$ white balls ($k = 0, 1, \dots, N$). An urn is chosen at random and n random drawings are made from it, the ball drawn being replaced each time.

Define

Event A : *all balls turn out to be red.*

Event B : *the $(n + 1)^{st}$ draw yields a red ball.*

1. Find $P(A \mid \text{Urn } k \text{ is chosen})$, ($k = 0, 1, \dots, N$).
2. Find $P(A)$, $P(AB)$ and $P(B \mid A)$.
3. Find an approximation to $P(B \mid A)$, using the fact that if N is large,

$$\frac{1}{N} \sum_{k=1}^N \left(\frac{k}{N} \right)^n \sim \int_0^1 x^n dx = \frac{1}{n+1}.$$

Exercise 3 – We consider a r.v. X random variable with the following distribution :

$$P(X = x) = (1 - p)^x p, \quad 0 < p < 1, \quad x \in \mathbb{N}.$$

1. Find the expectation of X .
2. Calculate the moment generating function and check the result obtained in question 1.
3. Let Y be another random variable with the same distribution, same parameter p , as X . Suppose X and Y are independent. Calculate the moment generating function of $X + Y$.
4. Let Z be a random variable with the following distribution :

$$P(Z = z) = \binom{n+z-1}{z} \theta^n (1-\theta)^z, \quad z \in \mathbb{N}, \quad 0 < \theta < 1.$$

- (a) Show that this is a probability.

Hint : Consider the Taylor's expansion of $1/(1-x)^n$

- (b) Express the moment generating function of Z .
5. Compare the distribution of $X + Y$ and Z .
 6. Identify the distribution of X , Y and Z , and write a conclusion.